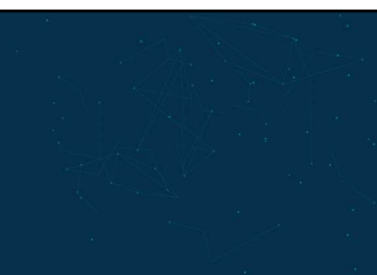


# Specialized and Fragile Systems

## CHAPTER 7



# 7.01 - Mobile Attacks

Episode **Mobile Attacks**  
title:

Objective: **3.5 Explain common attacks and vulnerabilities  
against specialized systems**

## MOBILE ATTACKS

- **Reverse engineering**
  - Decomposing steps in a process to determine weaknesses
  - Attackers use it to find vulnerabilities
  - Can be used to select appropriate controls

## MOBILE ATTACKS

- **Sandbox analysis**
  - Analyzing malware or exploit on isolated device or environment
  - Exploit is contained within the sandbox

## MOBILE ATTACKS

- **Spamming**
  - Type of DoS attack
  - Flooding channel with useless traffic to reduce the usability of useful messages

## MOBILE VULNERABILITIES

- **Insecure storage**
  - Unencrypted and easy for apps to access
- **Passcode vulnerabilities**
  - Weak and easy to guess
- **Certificate pinning**
  - Limiting certificates that devices or apps will accept

## MOBILE VULNERABILITIES

- **Using known vulnerable components**
  - Dependency vulnerabilities
    - Libraries or other included code
  - Patching fragmentation
    - Only patching part of a vulnerability or some systems/components

## MOBILE VULNERABILITIES

- **Execution of activities using root**
  - Potential for attacker to elevate permissions by leveraging an escape vulnerability
- **Over-reach permissions**
  - Allowing apps to do more than is necessary



## MOBILE VULNERABILITIES

- **Biometrics integrations**

- Any authentication method must be implemented securely and limited in its use

- **Business logic vulnerabilities**

- Vulnerable software risks data disclosure, alteration, and availability

## MOBILE TOOLS

- **Burp Suite** - <https://portswigger.net/>
- **Drozer** - <https://labs.f-secure.com/tools/drozer/>
- **Needle** - <https://github.com/FSecureLABS/needle>
- **Mobile Security Framework (MobSF)** - <https://mobsf.github.io/Mobile-Security-Framework-MobSF/>
- **Postman** - <https://www.appdome.com/how-to/dev-sec-tools/app-sec-release-automation/test-cicd-postman-api-app-security-projects/>
- **Ettercap** - <https://www.ettercap-project.org/>

## MOBILE TOOLS

- **Frida** - <https://frida.re/>
- **Objection** - <https://github.com/sensepost/objection>
- **Android SDK Tools** - <https://developer.android.com/studio>
- **Androzer (deprecated)** - <https://github.com/az0mb13/Androzer>
- **ApkX** - <https://github.com/b-mueller/apkx>
- **APK Studio** - <https://vaibhavpandey.com/apkstudio/>

## 7.02 -Virtual and Containerized Environment Attacks

Episode title: **Virtual and Containerized Environment Attacks**

Objective: **3.5 Explain common attacks and vulnerabilities against specialized systems**

## VIRTUAL ENVIRONMENT VULNERABILITIES

- **Virtual machine (VM) escape**
  - Accessing resources outside a VM's boundary
    - Memory, storage, processing

# VIRTUAL ENVIRONMENT VULNERABILITIES

- **Hypervisor vulnerabilities**

- DoS
- Code execution
- Memory corruption
- VM escape

## VIRTUAL ENVIRONMENT VULNERABILITIES

- **VM repository vulnerabilities**
  - Altering trusted images with malicious ones
  - Potentially easy way to trick users into running malware

## CONTAINERIZED ENVIRONMENT VULNERABILITIES

- **Vulnerabilities related to containerized workloads**
  - Workload remote access
  - Access to sensitive data and resources
  - Elevated privileges
  - Host escape



## 7.03 – Labtainers Lab (Industrial Control System)

Episode title: **Labtainers Lab (Industrial Control System)**

Objective: **3.5 Explain common attacks and vulnerabilities against specialized systems**